AURORA-ADAS Diagrams

AURORA-ADAS system architecture

```
+----------------------------------------------------------------+
|                           AURORA-ADAS                           |
+----------------------------------------------------------------+


   +-------------+      +-----------+      +-----------+
   | Perception  | ---> |   SLAM    | ---> |  Mapping  |
   | (ViT-DETR)  |      | (VIO/FG)  |      | (Semantic)|
   +-------------+      +-----------+      +-----------+
          |                    |                  |
          |                    |                  |
          v                    v                  v


   +------------------------------------------------------------+
   |                       State Estimation                     |
   |   (Pose, Velocity, Covariance, Semantic World Model)       |
   +------------------------------------------------------------+

          |
          v

   +----------------+      +----------------------+
   |      MPC       | ---> | RL Steering Prior    |
   | (Trajectory    |      | (Gaussian PDF, PPO/TD3)|
   | Optimization)  |      +----------------------+
   +----------------+                 |
          |                           |
          v                           v

+----------------------------------------------------------+
|                     Safety Supervisor                    |
|   (Collision Avoidance, Emergency Braking, Fail-Safe)    |
+----------------------------------------------------------+

       |
       v

+--------------------------+
|      UAV Actuators       |
| (Motors, Control Surfaces)|
+--------------------------+
```

Fig. 1. AURORA-ADAS system architecture showing the full autonomy pipeline from perception to control.

Perception pipeline

```
+------------------------------------------------------+
|                Perception Pipeline (ViT-DETR)        |
+------------------------------------------------------+

   +------------------+      +------------------+
   |   Raw RGB Image  | ---> | Patch Embedding  |
   |  (UAV Camera)    |      | (P×P tokens)     |
   +------------------+      +------------------+
                                    |
                                    v
                           +----------------------+
                           |  Transformer Encoder |
                           |  (Self-Attention)    |
                           +----------------------+
                                    |
                                    v
                           +----------------------+
                           |  Transformer Decoder |
                           |  (Object Queries)    |
                           +----------------------+
                                    |
                                    v
   +------------------+      +---------------------------+
   | Bounding Boxes   | ---> | Semantic Labels / Classes |
   | (DETR Outputs)   |      | (Obstacles, Roads, Crops) |
   +------------------+      +---------------------------+
                                    |
                                    v
                           +---------------------------+
                           |  Perception Outputs       |
                           |  (Detections + Features)  |
                           +---------------------------+
```

Fig. 2. ViT-DETR perception pipeline illustrating patch embedding, transformer encoding, decoding, and semantic output.

Factor-graph SLAM

```
+------------------------------------------------------+
|                 Factor-Graph SLAM                    |
+------------------------------------------------------+

   Time ---->

   x0          x1          x2          x3          x4
   |           |           |           |           |
   v           v           v           v           v

  [Pose0]----[Pose1]----[Pose2]----[Pose3]----[Pose4]
     |          |           |          |          |
     |          |           |          |          |
 IMU Factor  IMU Factor  IMU Factor  IMU Factor  IMU Factor
     |          |           |          |          |

   Cam0       Cam1        Cam2       Cam3       Cam4
    |          |           |          |          |
    v          v           v          v          v

  [Visual Factors] (Reprojection errors to landmarks)

   Loop Closure
   ------------
   Pose0  <---->  Pose3  (Loop-closure factor)

Optimization:
- Minimize sum of all factor residuals
- Output: Pose trajectory + map
```

Fig.3. Factor-graph SLAM with IMU factors, visual factors, and loop-closure constraints.

MPC flow diagram

```
+-------------------------------------------------------+
|               Model Predictive Control (MPC)          |
+-------------------------------------------------------+


   +------------------+
   | Current State xk |
   | (pose, vel, etc.)|
   +------------------+
             |
             v
   +---------------------------+
   | Prediction Model          |
   | (Dynamics over horizon N) |
   +---------------------------+
             |
             v
   +---------------------------+
   | Cost Function J           |
   | (tracking + control cost) |
   +---------------------------+
             |
             v
   +---------------------------+
   | Constraints               |
   | (speed, steering, safety) |
   +---------------------------+
             |
             v
   +---------------------------+
   | Optimization Solver       |
   | (argmin J subject to C)   |
   +---------------------------+
             |
             v
   +---------------------------+
   | Optimal Control Sequence  |
   | u* = {u0, u1, ..., uN}    |
   +---------------------------+
             |
             v
   +---------------------------+
   | Apply First Control u0    |
   +---------------------------+
             |
             v
   +---------------------------+
   | Next State x(k+1)         |
   | (Receding Horizon Update) |
   +---------------------------+
```

Fig.4. MPC flow diagram showing prediction, cost evaluation, constraint handling, and optimal control selection.

RL steering prior flowchart

```
+-------------------------------------------------------+
|          RL-Conditioned Gaussian Steering Prior        |
+-------------------------------------------------------+


   +----------------------------+
   | Environment / State s     |
   | (pose, obstacles, map)    |
   +----------------------------+
             |
             v
   +----------------------------+
   | RL Policy πθ(s)           |
   | (PPO / TD3)               |
   +----------------------------+
             |
             v
   +----------------------------+
   | Steering Parameters        |
   | μ(s), σ(s)                 |
   +----------------------------+
             |
             v
   +----------------------------+
   | Gaussian Steering PDF      |
   | δ ~ N(μ, σ²)               |
   +----------------------------+
             |
             v
   +----------------------------+
   | Sample / Select Steering   |
   | δ_rl                       |
   +----------------------------+
             |
             v
   +----------------------------+
   | Fusion with MPC Output     |
   | δ_final = f(δ_mpc, δ_rl)   |
   +----------------------------+
             |
             v
   +----------------------------+
   | Send to UAV Actuators      |
   +----------------------------+
```

Fig.5. RL-conditioned Gaussian steering prior combining PPO/TD3 outputs with MPC steering.

Semantic mapping pipeline

```
+-------------------------------------------------------+
|                 Semantic Mapping Pipeline             |
+-------------------------------------------------------+


   +-----------------------------+
   | Perception Outputs          |
   | (Detections, Classes)       |
   +-----------------------------+
              |
              v
   +-----------------------------+
   | 3D Projection / Raycasting|
   | (into world coordinates)  |
   +-----------------------------+
              |
              v
   +-----------------------------+
   | Voxel Grid / Map Cells      |
   +-----------------------------+
              |
              v
   +-----------------------------+
   | Bayesian Update             |
   | P(c | z) per voxel          |
   +-----------------------------+
              |
              v
   +-----------------------------+
   | Semantic World Model        |
   | (roads, obstacles, crops) |
   +-----------------------------+
              |
              v
   +-----------------------------+
   | Downstream Modules          |
   | (MPC, RL, Safety)           |
   +-----------------------------+
```

Fig.6. Semantic mapping pipeline using voxel grids and Bayesian updates to build a world model.

Full autonomy loop

```
+-----------------------------------------------------+
|                AURORA-ADAS Autonomy Loop            |
+-----------------------------------------------------+

   +----------------------------+
   | Onboard Sensors            |
   | (Camera, IMU, GNSS*, etc.)|
   +----------------------------+
              |
              v
   +----------------------------+
   | Perception (ViT-DETR)      |
   +----------------------------+
              |
              v
   +----------------------------+
   | VIO / SLAM (Factor-Graph) |
   +----------------------------+
              |
              v
   +----------------------------+
   | Semantic Mapping           |
   +----------------------------+
              |
              v
   +----------------------------+
   | State + World Model        |
   +----------------------------+
              |
              v
   +----------------------------+
   | MPC Trajectory Planning    |
   +----------------------------+
              |
              v
   +----------------------------+
   | RL Steering Prior          |
   +----------------------------+
              |
              v
   +----------------------------+
   | Safety Supervisor          |
   +----------------------------+
              |
              v
   +----------------------------+
   | UAV Actuators / Control    |
   +----------------------------+
              |
              v
   +----------------------------+
   | Next Sensor Measurements   |
   +----------------------------+
              |
              v
        (Loop repeats)
```

Fig.7. Full autonomy loop integrating sensing, perception, SLAM, mapping, planning, RL, safety, and actuation.